

```
import pandas as pd
import numpy as np
from sklearn.feature_extraction.text import TfidfVectorizer
import nltk
nltk.download('punkt_tab','stopword','wordnut')
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

```
data=pd.read_csv("/content/drive/MyDrive/Colab Notebooks/dataset1knn.csv")
```

data

	Message	Label	
0	Hi, how are you?	Ham	
1	Let's meet tomorrow.	Ham	
2	Call me when you reach home.	Ham	
3	Good Morning! Have a nice day.	Ham	
4	Can we meet at 5 PM today?	Ham	
5	Please send me the assignment.	Ham	
6	Happy Birthday! Have a wonderful day.	Ham	
7	I'll call you later.	Ham	
8	Congratulations! You won a lottery.	Spam	
9	Claim your free prize now.	Spam	
10	Win cash by clicking this link.	Spam	
11	Free recharge available today.	Spam	
12	Limited time offer. Click now!	Spam	
13	You have won a free iPhone.	Spam	
14	Buy one get one free.	Spam	
15	Urgent! Claim your reward immediately.	Spam	
16	Exclusive offer just for you.	Spam	
17	Click here to receive your cash prize.	Spam	
18	Your OTP is 456789.	Ham	
19	Thank you for your help.	Ham	

Next steps:

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```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 2 columns):
 #   Column  Non-Null Count  Dtype
---  -
 0   Message  20 non-null    object
 1   Label    20 non-null    object
dtypes: object(2)
memory usage: 452.0+ bytes
```

```
x=data['Message']
y=data['Label'].map({'Ham':'0','Spam':'1'})
```

```
from sklearn.feature_extraction.text import TfidfVectorizer
```

```
vector=TfidfVectorizer(stop_words='english')
```

```
x = data['Message'] # Reset x to the original text data  
x=vector.fit_transform(x)
```

```
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=42)
```

```
model1=KNeighborsClassifier(n_neighbors=3)
```

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```
model1.fit(x_train,y_train)
```

```
predictions = model1.predict(x_test)  
predictions
```

```
accuracy_score(y_test,predictions)
```

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```
ms = input()  
ms = vector.transform([ms])  
pred = model1.predict(ms)  
print(pred)  
if pred[0]=='1':  
    print('Spam')  
else:  
    print('Ham')
```

```
you won 1 lakh claim it  
['1']  
Spam
```

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